Life Protection · Public Health

JLU-SPH

2026 "Respiratory Detection Mini-Invention" Creative Design Workshop Activity Plan

Guarding Life with the Heart of Science

Responding to Public Health with the Power of Collaboration



School of Public Health, Jilin University

School of Public Health Jilin University

JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

Ⅰ、 Basic Information of the Activity

- Activity Name: "Innovative Design Workshop — Small Inventions for Respiratory Tract Detection"
- Activity Theme: Addressing key respiratory disease prevention challenges in schools, teams will use self-developed product concept sketches as demonstrations to guide secondary school students in designing portable detection inventions based on principles of synthetic biology
- Activity Type: ☑ Science Education ☐ Public Welfare ☐ Cultural Promotion ☐ Community Service ☐ Other:
- Organizer: JLU-SPH Team

Ⅱ、 Activity Objectives

1. Core Objective: To bridge the gap between textbook biological knowledge and the practical realities of school-based epidemic prevention. By using creative concept sketches as a practical medium, secondary students will gain insights into the implementation strategies of synthetic biology-based respiratory tract detection technologies, fostering technological innovation and nurturing awareness of solving local health challenges through scientific creativity.
2. Specific Goals:
   - ○ Knowledge/Cognitive Aspect: Understand the risks of cross-infection of respiratory diseases in schools, the operating principles of basic biological detection components, and the design concepts of portable, everyday detection products.
   - ○ Behavioral/Skill Aspect: Learn the full innovation process — "identify the problem, conceive a solution, and hand-draw product sketches" — and improve skills in visual design, logical communication, and group collaboration with peer evaluation.
   - ○ Emotional/Attitudinal Aspect: Challenge the stereotype that "scientific inventions only belong in laboratories," foster empathetic thinking, consider vulnerable populations (such as the elderly and young children) in design, and establish a humanistic concept of safeguarding others' health through technology.
3. Expected Outcomes:
   - ○ Coverage: 40–70 secondary school students
   - ○ Satisfaction Target: Over 90%
   - ○ Collection of student-originated invention concept sketches; compilation of a complete sample design manuscript package; preparation of a comprehensive classroom explanation script

Ⅲ、 Target Audience

- Core Audience: Secondary school students aged 12–18
- Audience Characteristics: Possess basic biological knowledge, have personally experienced seasonal respiratory infectious disease outbreaks in school; imaginative and fond of drawing and creative design; eager to convert scientific knowledge into practical solutions but lack guidance for innovation practice.

Ⅳ、 Activity Time and Location

- Date: June 2, 2026
- Total Duration: Approximately 40 minutes
- Venue: Multimedia classroom at partner secondary school

V、 Activity Content and Procedure

【Overview of Activity Procedure】

Phase	Time	Session Name	Detailed Content	Responsible Personnel	Required Materials / Key Filming Points
Phase	**10 min**	Venue Setup and Material Distribution	Adjust projector, display sample manuscripts, distribute drawing paper, colored pens, and evaluation forms	Logistics, Teaching Assistants	Complete sample sketches, drawing kits, evaluation forms
Main Activity Phase	**20 min**	Professional Explanation	Lecturer introduces key issues in campus epidemic prevention, presents four sets of original detection product sketches, and deconstructs the design logic	Lecturer	Demonstration manuscripts such as popular science PPTs (popular science PowerPoint presentations), color-changing pen pouches, and smart badges
Main Activity Phase	**40 min**	Independent Concept Sketch Drawing	Students, individually or in teams, creatively draw their invention design sketches focusing on respiratory tract detection and school health alert systems	All Teaching Assistants	Drawing paper, pencils, colored pens, sketchbooks
Main Activity Phase	**20 min**	Project Sharing and Peer Evaluation	Creators explain their design concepts on stage; teachers and peers provide feedback and suggestions for improvement	Main Lecturer	Projection equipment, reference evaluation forms for works
Closing Phase	**10 min**	Activity Summary and Questionnaire Collection	Lecturer summarizes key innovation concepts, distributes and collects activity feedback questionnaires	Main Coordinator, Teaching Assistants	Storage bags, opinion collection forms, small gifts

[Detailed Description of Each Stage]

- **Stage 1: Professional Explanation**
  - **Core Content:** The instructor introduces real-world pain points such as mass coughing during seasonal transitions on campus, increased risk of cross-transmission of respiratory viruses, and inconvenience of testing, thus raising key issues. Four sets of demonstration manuscripts are sequentially displayed: color-changing induction health pen cases, intelligent early warning campus ID badges, schematic diagrams of synthetic biology detection elements , and user interface mock-ups for a campus epidemic prevention mini-program. Each work's creative origin, detection principle, and usage scenario are explained in lay terms, fully demonstrating the thought process from identifying real-life problems to designing products, while avoiding overly complex technical jargon.
  - **Key Message:** Scientific inventions need not be confined to the laboratory; by integrating synthetic biology detection technology into everyday stationery or campus cards, we can help protect the respiratory health of those around us.

Stage 2: Free Creative Sketching

- **Core Content:** Students may choose to work individually or in teams, conceptualizing portable mini-inventions focused on two main directions: rapid detection of respiratory viruses and campus health early warning. They hand-draw the product appearance, internal structure, and usage functions. Teaching assistants provide rotating guidance, encouraging students to consider both practicality and the needs of susceptible groups, such as the elderly and young children.
  - **Key Message:** Turn the epidemic prevention challenges you observe on campus into your own exclusive detection invention by illustrating them.

- **Stage 3: Product Sharing and Peer Review Exchange**
  - **Specific Content:** Each group presents by projecting their hand-drawn sketches, explaining the product function, design inspirations, and how it addresses epidemic prevention challenges. Classmates in the audience use the evaluation reference sheet to offer optimization suggestions, while the instructor provides simultaneous advice on scientific optimization.
  - **Key Message:** Every sketch represents a youth's intention to safeguard campus respiratory health; exchanging ideas can perfect creativity.

VI、 Personnel Allocation and Division of Responsibilities

- **Overall Responsible Person, 1 person:** Coordinates with the school, oversees the entire process, and manages unexpected situations. Must possess strong organizational skills and be familiar with the synthetic biology detection project content.

- **Main Lecturer, 1 person:** Responsible for introducing the pain points, explaining the demonstration sketches, and summarizing the event. Should be well-versed in all demonstration design principles and adept at communicating complex scientific concepts in accessible language.
- **On-site Teaching Assistants, 3 persons:** Guide creative activities, distribute materials, and maintain classroom order. Must be patient and detail-oriented, with the ability to quickly organize students' innovative ideas.
- **Photography and Logistics, 2 persons:** Arrange equipment and take photos during the event

.

Ⅶ、Materials List

- Publicity and Material Category:
  - Science popularization leaflets
- Equipment Category:
  - Notebook computer, projector
  - Camera/High-definition mobile phone
- Teaching Materials Category:
  - Popular Science PPT (Popular Science PowerPoint Presentation)
  - Complete set of demonstration design manuscripts
  - Peer review reference sheet for works
  - Activity feedback questionnaires

Ⅷ、Risk Contingency Plan

- Personnel Risks: Students may accidentally injure their hands while using colored pens or experience sudden dizziness or discomfort. Administer treatment with a first aid kit; in severe cases, contact the school doctor and class teacher.
- Venue Risks: Temporary failure of multimedia equipment preventing projection. Switch to circulating paper versions of demonstration manuscripts; the instructor will hold and explain the manuscripts.
- Material Risks: Insufficient quantities of drawing paper or colored pens. Prepare spare drawing materials in advance to ensure all participants can create their designs.

Ⅸ、Effectiveness Evaluation

- Process Evaluation: Observe students' engagement and participation in creative activities and group sharing during class, and document the implementation process in real-time throughout the event.
- Satisfaction Evaluation: Collect completed paper feedback questionnaires and conduct random on-site student interviews; gather all responses immediately after the event.